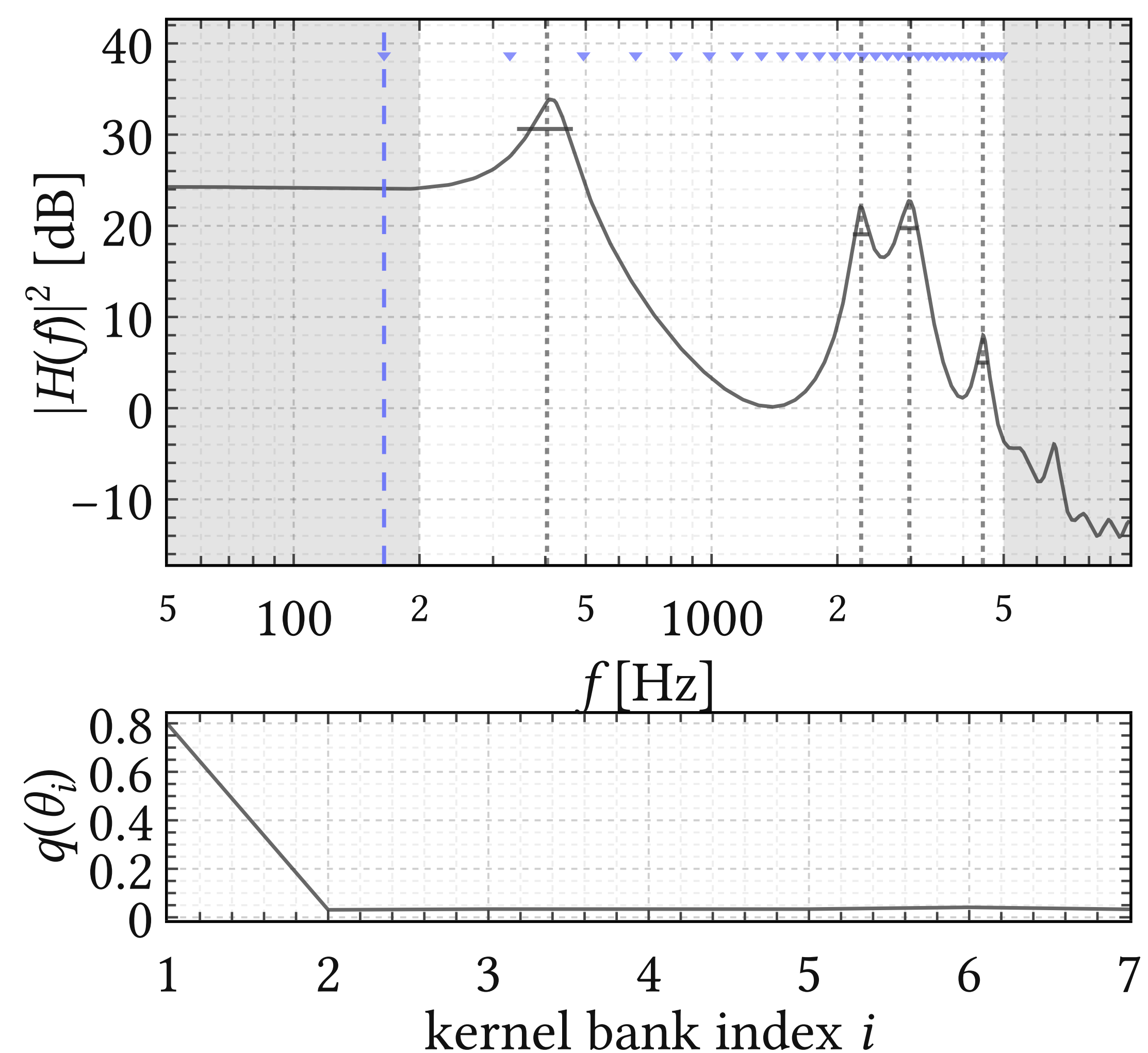
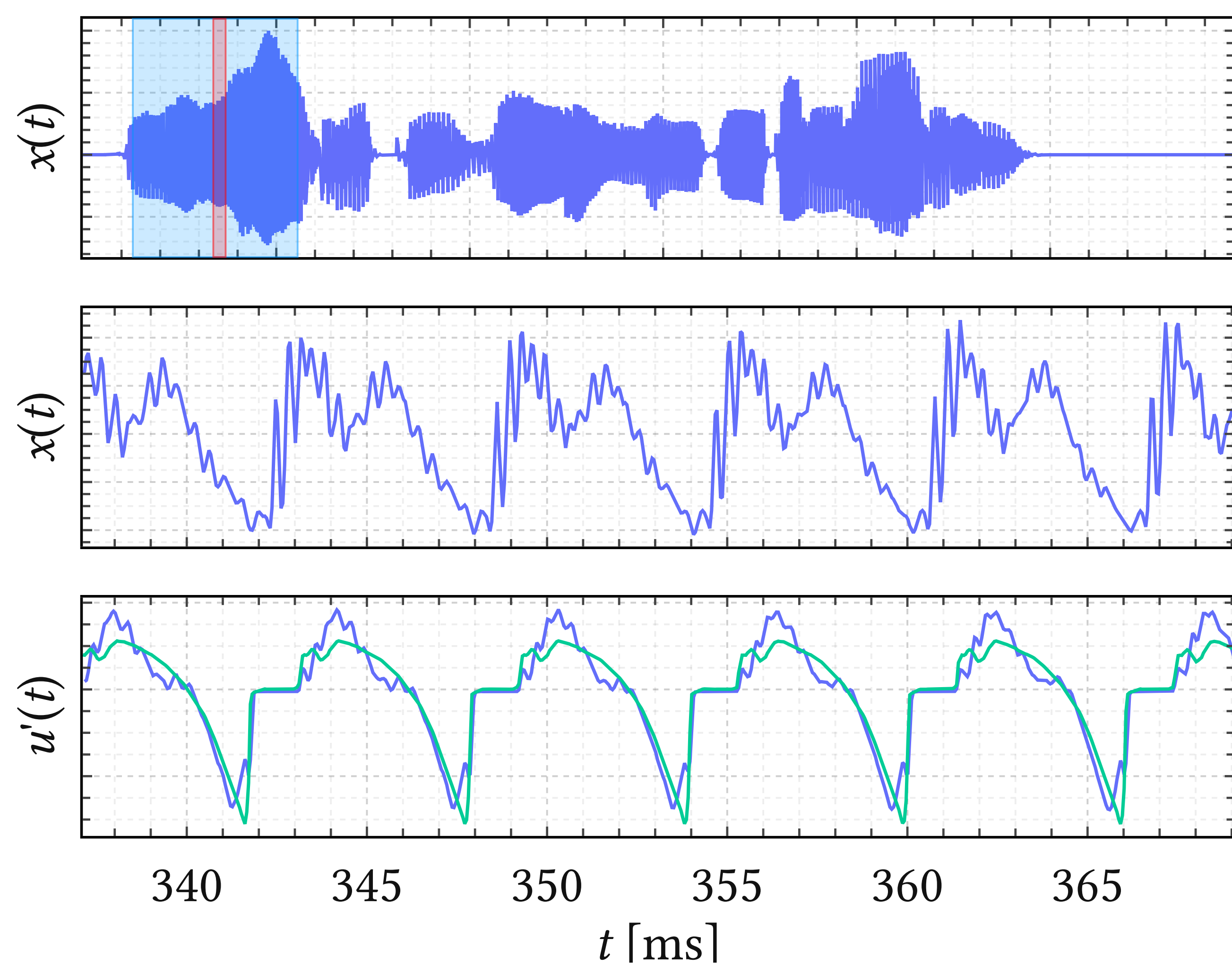


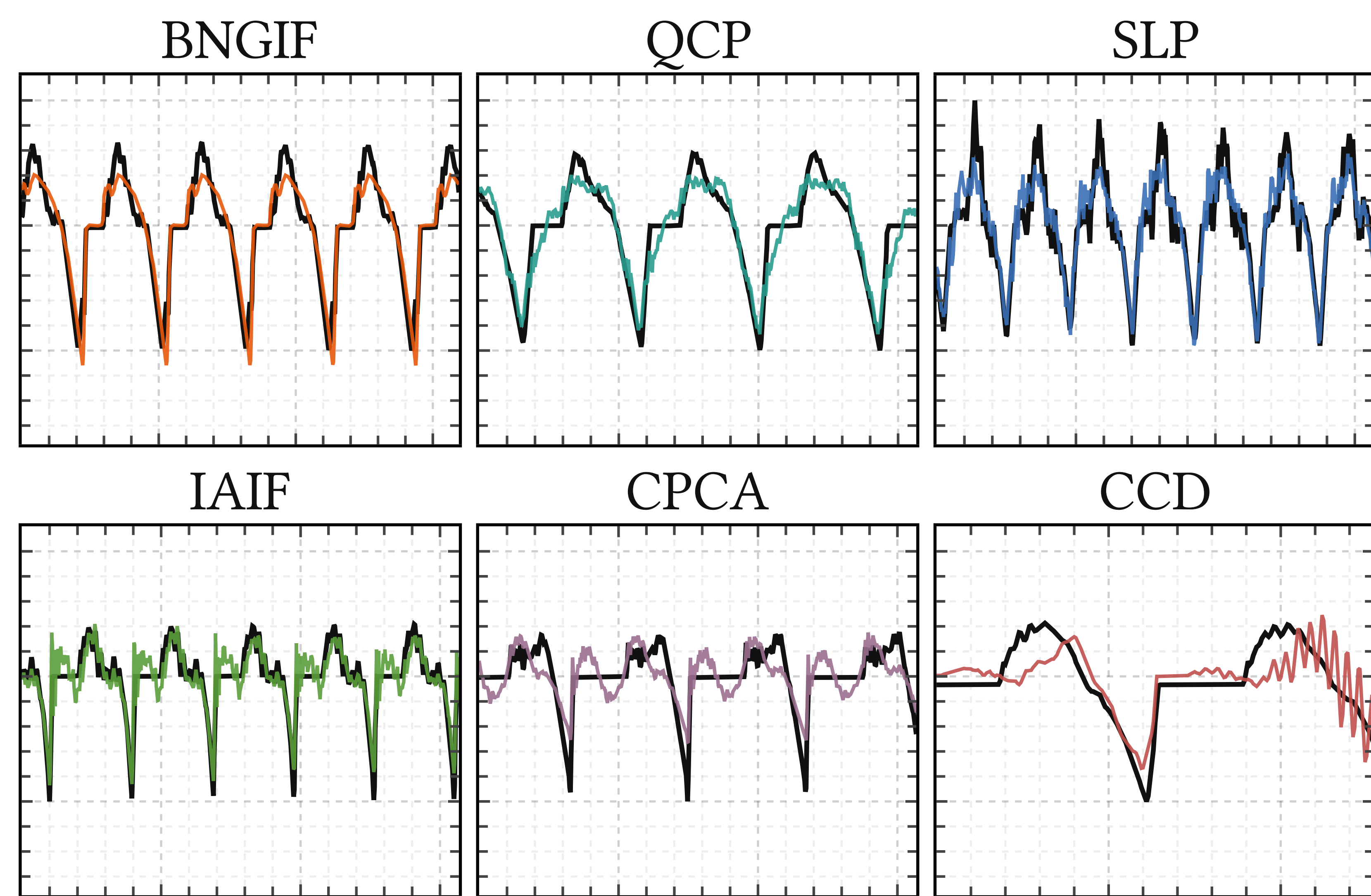
# PROBABILISTIC SPEECH DECONVOLUTION IN REAL TIME

## METHOD

One Bayesian model per 32 ms frame: a posterior over the glottal airflow (band), the vocal tract filter, and which prior shape explains the frame. One second of speech in 0.2 s on one GPU.



## BENCHMARK



|       | BNGIF        | QCP          | SLP          | IAIF         | CPCA         | CCD          |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|
| BNGIF | <b>0.935</b> | <.001        | <.001        | <.001        | <.001        | <.001        |
| QCP   | 1.000        | <b>0.910</b> | <.001        | <.001        | <.001        | <.001        |
| SLP   | 1.000        | 1.000        | <b>0.889</b> | <.001        | <.001        | <.001        |
| IAIF  | 1.000        | 1.000        | 1.000        | <b>0.886</b> | <.001        | <.001        |
| CPCA  | 1.000        | 1.000        | 1.000        | 1.000        | <b>0.759</b> | 0.004        |
| CCD   | 1.000        | 1.000        | 1.000        | 1.000        | 0.996        | <b>0.741</b> |

Paired Wilcoxon. Diagonal: median score; off-diagonal:  $p$ -value that the row beats the column. BNGIF beats each,  $p < .001$ .

True airflow (black) and six methods' estimates on the EGIFA benchmark.

## APPLICATIONS

### Hearing implants

Could pitch and formants *with error bars* improve sound coding?

### Clinics

Physiological voice biomarkers, not black-box features.

### Phonetics, forensics

Formant tracks 30% more accurate than Praat.

